KLE Technological University

Creating Value, Leveraging Knowledge

Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

Minor Project - 1 Report

on

Automated polymer bag handling system

By:

1. **Rohan Anvekar** USN:02FE21BEC074
2. **Rohit Hosalli** USN:02FE21BEC075
3. **Samruddhi V S** USN:02FE21BEC083
4. **Ujwala V T** USN:02FE21BEC105

Semester: VI, 2023-2024

Under the Guidance of

Dr. Prabhakar Manage

KLE Technological University,
Dr. M. S. Sheshgiri College of Engineering and Technology
BELAGAVI-590 008
2023-2024

Dr. M. S. Sheshgiri Campus, Belagavi

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled " **Automated polymer bag handling system** " is a bonafide work carried out by the student team of **"Rohan Anvekar 02fe21bec074, Rohit Hosalli 02fe21bec075, Samruddhi Shindolkar 02fe21bec083, Ujwala V T 02fe21bec105"**. The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for B.E. (VI Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2023-2024.

Dr. Prabhakar Manage	**Dr. Dattaprasad A. Torse**	**Dr. S. F. Patil**
Guide	**Head of Department**	**Principal**

External Viva:

Name of Examiners **Signature with date**

1.
2.

ACKNOWLEDGMENT

I want to thank dr.Prabhakar Manage for helping me a lot with our project on "Automated polymer bag handling systemg". He gave me great advice, supported us all the way,and worked really hard. His teaching and guidance made our project better. Prof. Managewas not just a teacher but also a mentor, making our learning more practical. He was patient and encouraging when things got tough, and his help was not just about the technical stuff but also about working well together as a team. I'm really thankful for his support, and our project wouldn't be as good without him. I extend my sincere gratitude to our Head of Department, Dr. Dattaprasad A. Torse, and our college Principal, Dr. S. F. Patil, for their invaluable support during our project on "Crop Health Assessment through Image Processing". Dr.Torse provided crucial guidance and encouragement, significantly improving our project with insightful advice. I appreciate the conducive learning environment fostered by both Dr. S. F. Patil and Dr. Dattaprasad A. Torse , which allowed us to apply theoretical knowledge to real-life situations. Their leadership has been instrumental in shaping our academic journey, and I am thankful for their unwavering support and contributions to our project's success.

-The project team

ABSTRACT

The automated polymer bag handling system revolutionizes industrial packaging processes with its cutting-edge technology and sophisticated design. Featuring robotic arms equipped with specialized grippers, the system seamlessly integrates into production lines, efficiently picking up polymer bags and placing them onto conveyor belts with unparalleled precision. Leveraging advanced sensors and machine vision, the system ensures accurate bag detection, positioning, and alignment, optimizing workflow efficiency and minimizing errors. With intuitive software interfaces enabling real-time monitoring and control, operators can easily oversee operations and make adjustments as needed, enhancing productivity and reliability in manufacturing operations.

Designed for scalability and adaptability, the system offers a versatile solution to meet the diverse needs of different production environments. Its modular architecture allows for easy customization to accommodate various bag sizes, shapes, and materials, ensuring seamless integration into existing workflows. By automating bag handling tasks, the system reduces labor costs, minimizes downtime, and improves overall operational efficiency, making it a valuable asset for industries reliant on polymer bag packaging. With its innovative features and user-friendly interface, the automated polymer bag handling system sets new standards for efficiency, reliability, and performance in industrial packaging applications.

Contents

List of Tables

List of Figures

Chapter 1

Introduction

In contemporary industrial manufacturing, effective product handling and packaging are crucial for efficiency and meeting production goals. Polymer bags, used widely across industries, are essential for protecting and transporting goods. Manual handling of these bags is labor-intensive, time-consuming, and prone to errors, creating inefficiencies. Innovative solutions for streamlining bag handling are increasingly needed to optimize workflow and remain competitive.

Automated polymer bag handling systems, utilizing robotics, sensors, and control algorithms, address these challenges by automating the process, from loading to placement. These systems improve workflow, enhance safety, and ensure consistency in packaging, revolutionizing bag handling with significant efficiency and productivity gains.

Manufacturing companies adopting these automated systems benefit from reduced manual labor, minimized errors, enhanced safety, optimized resource use, and greater flexibility to meet growing demands. This strategic investment boosts their competitive edge and drives sustainable growth in today's dynamic market.

1.1 Motivation

The drive to develop and implement automated polymer bag handling systems stems from the quest for operational excellence and a competitive edge in industrial manufacturing. Automating bag handling processes brings numerous benefits, enhancing efficiency, productivity, and profitability.

Automation replaces labor-intensive manual tasks, reducing manpower reliance and minimizing human errors. This boosts production efficiency and ensures consistent quality and accuracy in packaging. Additionally, automated systems enhance workplace safety by reducing injury risks associated with manual handling, leveraging robotics and advanced sensors for precise and reliable task execution.

Adopting automated systems also enables companies to respond to market demands with greater agility and scalability. These systems allow quick adjustments to production volumes, ensuring responsiveness to market dynamics.

1.2 Objectives

- Labor Reduction:Utilize automation to perform repetitive and labor-intensive tasks, reducing the reliance on manual labor.
- Scalability and Flexibility: Implement modular systems that can be easily expanded or reconfigured to meet changing production demands.
- Efficiency Enhancement: Use real-time monitoring and feedback systems to quickly identify and address inefficiencies.
- Improved Product Quality: Ensure gentle handling of polymer bags to prevent damage to the contents.Maintain a high level of consistency in handling, reducing the risk of product contamination or damage.
- Reliability and Maintenance: Design systems to be durable and reliable, minimizing downtime and maintenance requirements.Ensure the system is easy to maintain with readily available parts and clear maintenance procedures.

1.3 Literature survey

Table 1.1: Literature Review

SI.No	**Paper Title**	**Objective**	**Algorithm used**	**Limitations**
1	[4]	Assessment of Vacuum Generator Technology and Investigation of Shock Absorber Usage.	A blob-detection algorithm and uses an end-effector pressure sensor to measure grip success.	Accurately identify objects in specific conditions and assess gripping success with potential inaccuracies.
2	[3]	To evaluate the effectiveness of a vacuum lift system in reducing spinal loads during baggage handling tasks	Optical motion capture markers were placed head-to-toe to capture body kinematics, aligned with OptiTrack's baseline placement locations.	Laboratory-Based Task Simulation and Single Type of Bag.
3	[1]	Use a pick-and-place system to differentiate bag colors, check dimensions, and position bags accurately.	Electronic sensing, embedded control system, and vacuum suction end-effector.	Sensitivity to Environmental Conditions and Limited Flexibility.
4	[2]	Emphasizing Customization for Specific Applications and evaluation of vacuum gripper performance.	The operation of the vacuum gripper, elucidating how it generates a vacuum through high-speed airflow to grip objects securely.	Surface Suitability and Inaccuracies in the assessment of gripping success.

1.4 Problem statement

The reliance on manual handling methods for polymer bags in agricultural supply chains presents inefficiencies, labor constraints, and potential errors. Developing an automated polymer bag handling system seeks to address these challenges by improving efficiency, reducing labor dependence, and enhancing overall operational effectiveness.

1.5 Application in Societal Context

- Improved Working Conditions: Automation reduces the physical strain on workers, minimizing the risk of injuries related to heavy lifting and repetitive tasks. This leads to better health and safety standards in the workplace.
- Economic Growth: Increased efficiency in the agricultural supply chain can lower production costs and lead to lower prices for consumers. This can boost the economy by making agricultural products more accessible and affordable.
- Rural Development: Implementing advanced technologies in rural areas can stimulate development by attracting investments and improving infrastructure. This can lead to better living standards and economic opportunities for rural communities.
- Environmental Benefits: More efficient handling systems can reduce waste and energy consumption, contributing to environmental sustainability. This aligns with broader societal goals of reducing the ecological footprint of agricultural practices.
- Increased Productivity: By automating labor-intensive tasks, productivity can be significantly increased, leading to higher output and the ability to meet growing food demands. This is crucial for food security and addressing hunger in various communities.
- Enhanced Food Safety: Automation can help maintain higher standards of hygiene and reduce contamination risks, ensuring safer food handling and processing. This benefits public health and consumer trust.
- Economic Equity: By lowering production costs and improving efficiency, small and medium-sized farms can compete more effectively with larger operations, promoting fairer economic opportunities within the agricultural sector.

1.6 Project Planning and bill of materials

- Project initiation -

  Objectives – The project aims to enhance the production process of polymer bags by increasing efficiency, reducing labor costs, and improving safety standards. This involves optimizing workflows to achieve higher output using the same or fewer resources, potentially through task automation and streamlined processes. Concurrently, the focus is on minimizing expenses associated with labor by introducing automation where feasible or reorganizing tasks to require fewer labor hours. Safety protocols will be enhanced to mitigate risks to employees' health and well-being, potentially through the implementation of proper training programs and the introduction of automated systems to handle hazardous tasks. By addressing these key areas, the project seeks to not only boost productivity but also create a safer and more cost-effective working environment.

- Planning phase –

  Currently focusing on

1 Defining the Requirements: Identify system functionalities, user interface, performance, and safety considerations.

2 Select Hardware: Choose ESP32 board, stepper motors, drivers, and power supplies based on project needs.

3 Design Circuit: Wire components ensuring correct connections and power distribution.

4 Software Design: Decompose functionality, define data structures, and develop algorithms for motor control.

5 Implementation: Write modular, well-commented code focusing on readability and error handling.

6 Testing: Test individual components, integration, and system functionality for reliability.

7 Documentation: Document hardware connections, software architecture, and usage instructions.

8 Deployment and Maintenance: Deploy the system, provide user training, and ensure ongoing maintenance and updates for longevity.

- Simulation testing -

1 Circuit Simulation: Implement and wire the components according to the hardware design.

2 Software Integration: Transfer and test the code on the simulated Arduino board.

3 Functional and Performance Testing: Evaluate motor control, serial communication, and error handling.

4 Documentation and Iteration: Document simulation setup, test results, and iterate based on feedback for enhanced system reliability.

  Gather feedback and make necessary adjustments

1.7 Organization of the report

- Introduction: - Brief overview and rationale for the project.
- Literature Review: - Summary of existing automated bag handling algorithms and applications.
- Objectives: - Clear definition of project goals and objectives.
- Methodology: - Explanation of the algorithm design process, and optimization strategies.
- System Architecture: - Overview of the overall system architecture.
- Algorithm Methodology: - Detailed explanation of algorithmic design principles, emphasizing optimization considerations.
- Simulation Testing: - Overview of testing procedures and presentation of simulation results.
- Ethical Considerations: - Discussion of ethical considerations related to data security.
- Results and Discussion: - Summary of key findings, comparative analysis, and optimization impact.

Chapter 2

System design

In this Chapter, we list out the interfaces. Modules:

- System Overview: The system controls two stepper motors (Motor X and Motor Y) using an ESP32 microcontroller, with user interaction facilitated through the Serial Monitor.
- Hardware Components: Utilizes an ESP32 board, stepper motors, stepper motor drivers, and power supplies for operation.
- Software Components: The Arduino sketch, written in C, manages motor movement based on commands received via serial communication.
- Functional Components: Includes a command parser, motor control logic, and error handling to ensure accurate and reliable motor control.
- User Input: Commands are sent via the Serial Monitor, specifying the direction and number of steps for each motor.
- ESP32 Output: The ESP32 processes commands and controls the stepper motors accordingly, providing feedback via the Serial Monitor.
- System Architecture: Comprises input, processing, and output layers, handling command parsing, motor control, and user feedback.
- System Constraints: Considerations include hardware limitations, such as power and communication speed, ensuring reliable operation.
- Assumptions: Assumes a stable power supply and reliable serial connection between the Arduino and the computer running the Serial Monitor.
- System Diagram: A visual representation illustrates the components, interactions, and flow of data within the stepper motor control system.

2.1 Functional block diagram

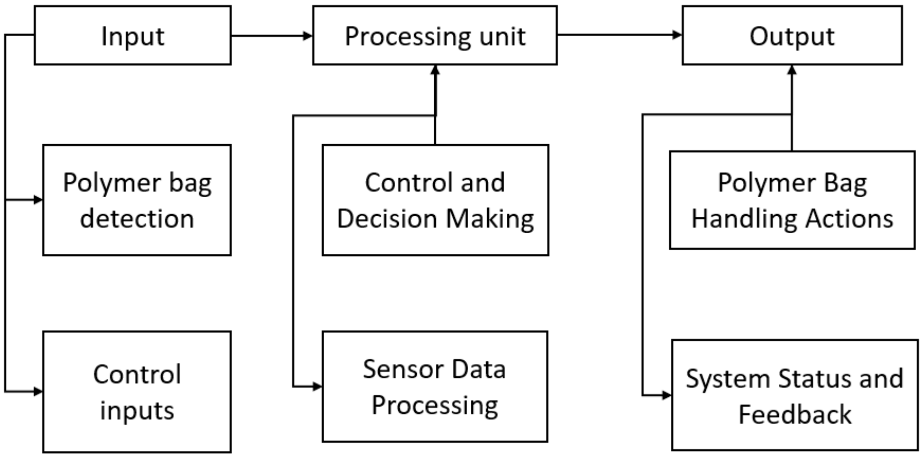

Figure 2.1: functional block daigram

2.2 Design alternatives

- Alternative Microcontroller Platforms

1 Raspberry Pi:

More powerful, can run a full operating system, capable of handling complex tasks and additional peripherals. More expensive, higher power consumption, requires more complex setup and programming.Suitable for applications requiring additional processing power, complex user interfaces, or connectivity options like Wi-Fi or Bluetooth.

2 ESP32:

Built-in Wi-Fi and Bluetooth, more GPIO pins than basic Arduino boards, higher processing power. More complex to program compared to Arduino, may require additional setup. Ideal for IoT applications or projects needing wireless connectivity.

- Identifying the multiple solutions

1 Bag-loading conveyor belt

Automated Transport: A motor-driven continuous loop belt moves bags placed on it to designated loading points.

Controlled Operation: Sensors and control systems manage the speed and positioning of the belt.

System Integration: Often integrates with pick-and-place robots, enhancing productivity and reducing manual labor.

2 Mechanical gripper

Design: Mechanical grippers with rigid or flexible fingers are integrated into automated bag handling systems to grasp and hold bags securely.

Actuation: Gripper actuation, often pneumatic or electric, enables the controlled opening and closing of the gripper to pick up and release bags as required.

Integration: These grippers are seamlessly integrated into robotic systems or conveyor belts, facilitating the automated handling of bags in processes such as loading, sorting, and packaging.

3 Vacuum lifter

Design: Vacuum lifters are designed with a suction pad or cups that adhere to the surface of polymer bags through negative pressure, creating a secure grip.

Actuation: The vacuum lifter is connected to a vacuum pump that generates the suction force. When activated, the pump creates a vacuum between the lifter's pad/cups and the bag's surface, allowing it to lift and move the bag.

Application:They efficiently lift and transport bags from one location to another, minimizing manual labor and ensuring gentle handling of fragile polymer materials.

2.3 Final design

We select one of the optimal solutions based on its working and ease of the implementation.

- Final Design:

1 ESP32

Wireless Communication: The ESP32 can facilitate wireless communication between the automated bag handling system components, such as grippers or vacuum lifters, and the central control system.

Automation: They enable the grippers or vacuum lifters to move along predefined axes, allowing for accurate positioning and manipulation of bags during the handling process. The ESP32 microcontroller can interface with these actuators to control their operation, enabling coordinated movements and efficient operation of the bag handling system.

2 Vacuum lifter

Vacuum Generation: The vacuum lifter contains a vacuum pump or generator that creates suction. This suction force is used to hold the material securely against the lifter.

Attachment to Material: The vacuum cups are placed against the surface of the material to be lifted. When the vacuum pump is activated, air is removed from the space between the cup and the material, creating a vacuum that holds the material firmly in place.

Release Mechanism: After the material has been positioned or placed at its destination, the vacuum lifter can release the material by either shutting off the vacuum pump or by actively releasing the vacuum pressure, allowing the material to be gently lowered or released.

Chapter 3

Implementation details

3.1 Specifications and final system architecture

Specifications:

- Integration of Motor Control Functions: Integrate motor control functions into the Arduino sketch to perform precise control of servo, DC, and stepper motors.Define clear steps for controlling each motor type based on user commands received through serial communication.
- Comprehensive Testing: Conduct testing to validate the functionality and performance of motor control operations under various scenarios.Include simulation and real-world testing to ensure accurate control of servo, DC, and stepper motors based on user commands.Verify reliability and safety aspects to prevent motor overheating, overcurrent, or mechanical damage.
- Arduino Sketch Documentation: Document the Arduino sketch with clear comments, describing the purpose and functionality of each section of code. Provide explanations for motor control functions, serial communication handling, and overall system operation.
- System Architecture Documentation: Define the system architecture with diagrams illustrating the connections between Arduino, motors, and any additional components. Include descriptions of hardware components, software modules, and their interactions within the system.

3.1.1 Model specifications

- Servo motor contol :

  The servo motor is used to control the position of an attachment mechanism.

  Model: mg90 servo motor.

  Specifications: Typically operates within a range of 0 to 180 degrees

  Purpose: Controls the position of the arm or attachment mechanism of the vacuum lifter

  Connection: Attached to pin 10 of the ESP32 board

- Dc motor via relay :

  The relay-controlled DC motor can represent the vacuum pump or generator in a vacuum lifter system.

  When the relay is turned on, it activates the DC motor, which in a vacuum lifter would power the vacuum pump.

  Model: 6-12V DC water pump.

  Specifications: Provides suction force through a vacuum pump or generator

  Purpose: Creates suction through the vacuum cup(s) to grip the material being lifted

  Connection: Controlled by a relay connected to pin 11 of the ESP32 board

- Stepper Motors:

  Model: nema 17 stepper motors for Motor X and Motor Y

  Specifications: Capable of precise movement in both directions

  Purpose: Controls the movement of the vacuum lifter platform in the X and Y directions

  Connection: Connected to specific pins for step, direction, and enable signals (stepPinX, dirPinX, enablePinX for Motor X; stepPinY, dirPinY, enablePinY for Motor Y)

- Control Interface:

  Serial Communication: Allows external control of the vacuum lifter via serial commands

  Command Structure: Commands are sent over serial communication to control servo position, activate/deactivate the vacuum pump, and move the stepper motors in the X and Y directions

3.1.2 Flow chart

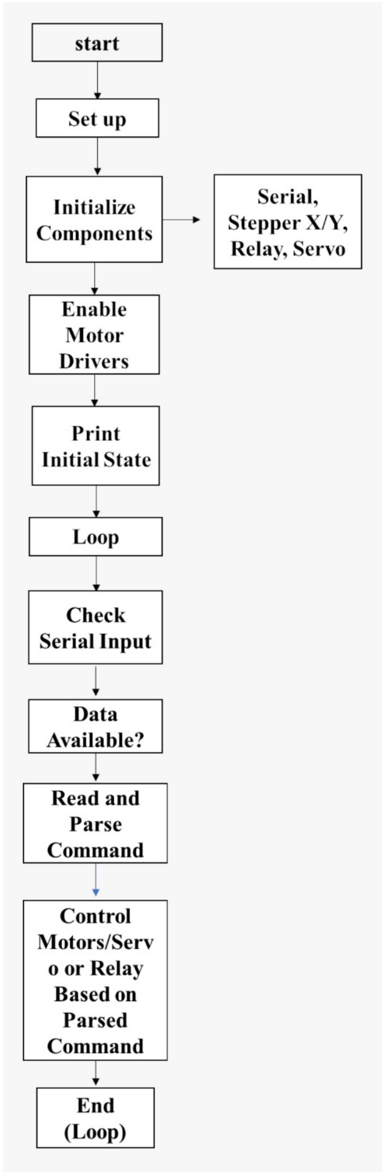

Figure 3.1: flow chart of our mechanism

Chapter 4

Optimization

4.1 Introduction to optimization

Optimization involves enhancing system performance by improving its components. For the pick-and-place machine, an advanced microcontroller will control the X, Y, and Z axes and the vacuum pump, operated via a Bluetooth app. This integration ensures higher accuracy and precision in operations.

Centralizing all circuit connections will simplify the design and improve the machine's aesthetics. This approach reduces complexity, enhances signal integrity, and minimizes failure points. Overall, these optimizations will result in a more efficient, reliable, and user-friendly machine.

4.2 Types of Optimization

1 Bluetooth Optimization

Replacing the serial monitor with a Bluetooth app for controlling the machine Enhances the user interface, making it more intuitive and accessible Allows for remote operation, providing greater flexibility and ease of use Reduces the complexity of physical connections, minimizing potential points of failure

2 Code Optimization

Writing raw code instead of relying on driver libraries Increases execution speed by eliminating unnecessary overhead from external drivers Enhances reliability and stability of the system by reducing dependencies on third-party code

4.3 Discussion on Optimization

Moved from Serial Monitor to Bluetooth App Enhanced user interface by replacing the serial monitor with a Bluetooth app Improved convenience and accessibility for remote operation and monitoring Allowed for more intuitive control and better user experience Reduced the complexity of physical connections, minimizing potential points of failure

4.4 Code optimization

Code optimization involved writing raw code without relying on external driver libraries Reduced system overhead, resulting in faster execution and more efficient operation Enhanced reliability by minimizing potential points of failure associated with driver dependencies Simplified maintenance and updates, as the system becomes less dependent on third-party libraries

Chapter 5

Results and discussions

5.1 Result Analysis

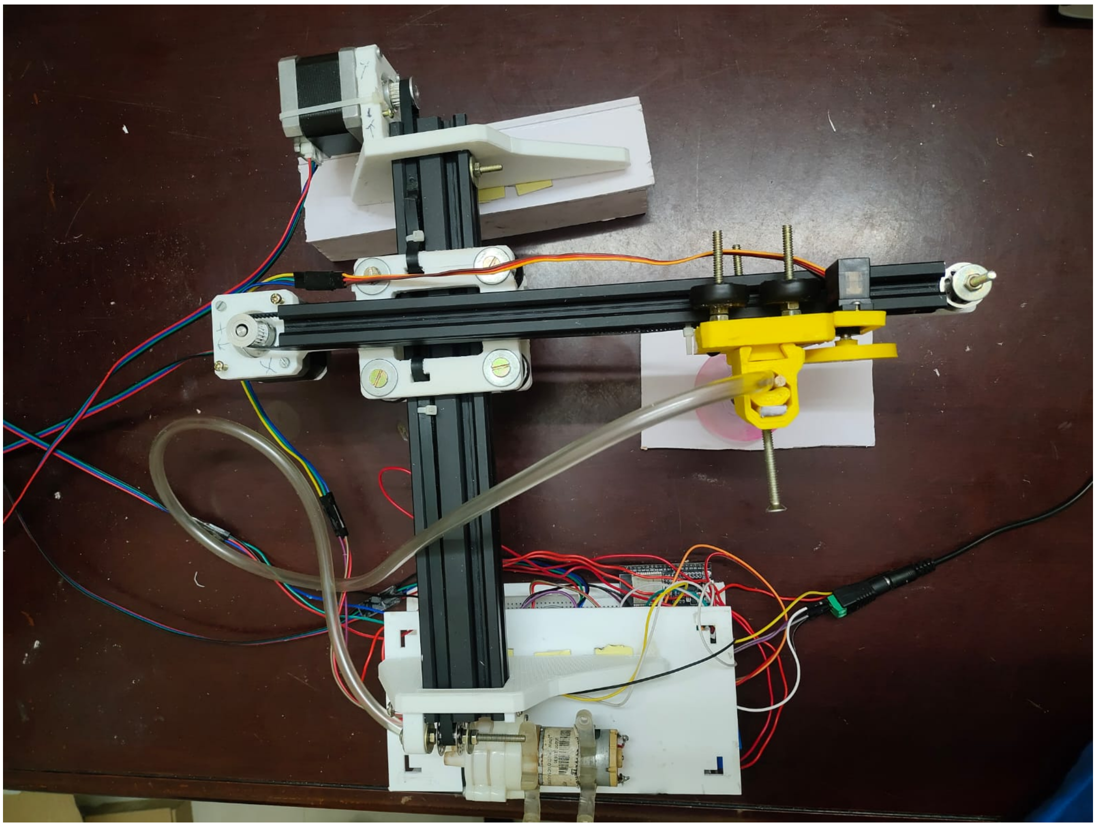

Figure 5.1: result

Chapter 6

Conclusions and future scope

6.1 Conclusion

The deployment of automated polymer bag handling systems represents a significant leap forward in the agricultural and industrial sectors, where the handling of bulk materials in polymer bags is a common yet labor-intensive task. These systems have demonstrated substantial benefits, including increased efficiency, reduced labor costs, improved safety, and enhanced accuracy in the handling and storage of polymer bags. By leveraging technologies such as robotic arms, conveyor systems, and automated sorting mechanisms, these systems streamline operations, minimize human error, and ensure consistent handling processes.

Automated polymer bag handling systems also contribute to better inventory management through precise tracking and monitoring, which helps in optimizing storage space and reducing waste. Despite the substantial initial investment required for installation and setup, the long-term gains in productivity, cost savings, and operational efficiency make these systems a worthwhile investment for many businesses.

6.2 Future scope

- Customization and Scalability:

  Developing customizable and scalable solutions that can be tailored to specific industry needs and varying operational scales will enhance the versatility of automated gunny bag handling systems. Modular designs can allow for easy upgrades and expansions as business needs grow.

- Safety Features:

  Incorporate safety features such as emergency stop buttons, overcurrent protection, and thermal monitoring to protect the system and operators from potential hazards.

- Real-time Monitoring and Diagnostics:

  Develop a comprehensive monitoring system that provides real-time data on the status and performance of the bag handling system. This could include predictive maintenance features to identify potential issues before they lead to downtime.

- Global Expansion and Adoption

  As automated polymer bag handling systems prove their value, their adoption will expand globally. Developing countries with growing manufacturing sectors will increasingly invest in these technologies to boost their competitiveness and meet international standards.

Bibliography

[1] Awais Farooqi, Nukman Bin Yusoff, and Lim Siong Chung. Automatic pick-and-place packaging system with vacuum lifter. In *IOP Conference Series: Materials Science and Engineering*, volume 697, page 012008. IOP Publishing, 2019.

[2] AK Jaiswal and B Kumar. Vacuum cup grippers for material handling in industry. *Int. J. Innov. Sci. Eng. Technol*, 4(6):187–194, 2017.

[3] Ming-Lun Lu, Jonathan S Dufour, Eric B Weston, and William S Marras. Effectiveness of a vacuum lifting system in reducing spinal load during airline baggage handling. *Applied ergonomics*, 70:247–252, 2018.

[4] Emmanouil Papadakis, Fredy Raptopoulos, Maria Koskinopoulou, and Michail Maniadakis. On the use of vacuum technology for applied robotic systems. In *2020 6th International Conference on Mechatronics and Robotics Engineering (ICMRE)*, pages 73–77. IEEE, 2020.